Embedded Computation in the Incompleteness of Observation Framework

From Evolution to BQP and the P vs NP Question

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Abstract

The Observational Incompleteness (OI) framework reproduces the Standard Model's gauge structure as a forced consequence of a cubic-lattice commitment — its dimension argued by self-consistency filters, its cubic structure empirically anchored by the observed SU(3) — starting from a single definition of embedded observation [1, 2] and traces the structural chain from (S, φ) through chemistry to biological evolution [3]. This paper continues the chain from evolution through computation. The transition from evolved biological observers to artificial computational ones admits a uniform treatment in the framework's terms: an embedded observer with general intelligence can construct semiconductor-based universal computers via a structural chain (band theory, Group IV semiconductors, p-n junctions, NAND completeness), and the resulting artificial observers are subject to the same partition-level limits as their biological precursors. The Gödel-Turing-OI incompleteness family applies at every level of recursive self-improvement.

The framework's computational content is organized by a *conditional BQP model theorem*, stated as the statement of record in §4 of this paper and developed at chapter length in [1, Ch. 14]: under an explicit computational model — a uniform efficient encoding of substratum configurations into qubits; efficient state preparation and control; a clock model; efficient measurement; explicit precision accounting; noise assumptions under which fault tolerance applies with polynomial overhead; and operational closure — all admissible operations available to the observer within the model are standard quantum instruments (CPTP maps for unconditioned evolution, CP trace-nonincreasing branches for conditioned outcomes), with no additional nonlinear, post-quantum, oracle, or hidden-sector-access operation, and with conditioned branches and postselected preparations charged their physical success-probability/repetition cost (no free postselection: PostBQP = PP) — the class accessible to an embedded observer satisfying C1–C4 is bounded-error quantum polynomial time, with a lower bound (BQP available in the model) and an upper bound (BQP not exceeded — given operational closure). CPTP representability of the visible dynamics supplies none of those model hypotheses by itself, so no unconditional characterization is claimed. Under a global-coverage premise — every physically realizable computation implemented by a system whose admissible operations

satisfy the same operational-closure and efficiency hypotheses — the Extended Church-Turing Thesis holds in its quantum-modified form as a conditional result rather than an empirical conjecture. The conjectured BPP–BQP separation is reframed as the cost of marginal-statistics observation rather than as evidence of fundamental quantum-ness.

On the P vs NP question the framework is *structurally silent*. We develop this position in detail: the framework's empirical content (the BQP ceiling) is consistent with either answer; the silence is not a chosen agnosticism but a structural feature of the framework's axiom type; physical-substrate axioms about finite structures do not entail asymptotic statements about algorithmic complexity classes without a bridging premise that the candidate axioms cannot principled-ly supply. This silence fits a broader pattern across the Gödel-Turing-OI incompleteness family: each member establishes a structural boundary on access (provability, computability, embedded observability, explanation) and is silent on the internal organization of what falls within that boundary. P vs NP lies in the internal structure of the computable and is therefore outside the locus of contribution of any structural-incompleteness framework. The OI framework's position is stronger than Aaronson's NP-completeness physical-principle approach [4] at the BQP boundary (where the framework specifies the ceiling exactly) and weaker at the P vs NP boundary (where the framework declines to stipulate). We propose three condition-based research directions where the framework's specific substratum structure does provide content for complexity-theoretic refinements *within* the BQP ceiling: geometric structure on the substratum lattice, task-specific quantum advantage and problem globality (with a refined conjecture that BQP advantage is maximized for problems whose hidden structure matches the framework's gauge group), and gauge equivalence as a possible physical account of polynomial-time reducibility. These are framework-relative conjectures inviting investigation, not framework predictions.

Status — conjectural extension. This paper applies the core framework to quantum computation and complexity, and is offered as a *conjecture*, not an established result: the central claim — that the framework's emergent dynamics realizes the complexity class BQP — is a theoretical conjecture rather than a proven correspondence; it would be supported by a rigorous derivation from the finite substratum to BQP and undercut if that dynamics proves efficiently classically simulable or fails to reproduce known quantum-complexity separations. It carries no evidential weight for the core framework — which stands or falls independently on its physics — and should be read as a structural possibility to be tested, not as support for the framework.

1. Introduction

The structural chain from (S, φ) through chemistry to biological evolution is developed in the companion paper [3]. The thermal window, organic chemistry, autocatalytic networks, template replication, the origin of life as a molecular C1–C4 system, and Darwinian evolution — each step is consequence-tracing in the framework's material-structural mode. The chain has a natural terminus at biological evolution: from there, the next link (information processing $\to$ general intelligence $\to$ artificial computation) introduces a qualitatively different kind of consequence, in the framework's *informational-structural* mode.

This paper develops that next link. The transition from evolved biological observers to artificial computational ones is itself a structural chain — fitness pressure favors information-processing organisms, neural computation is materially possible given the framework's chemistry, and once general intelligence exists, semiconductor-based universal computation follows from the derived periodic table and the scale hierarchy. AI is an embedded observer building another embedded observer, with the partition-level limits applying to both. The Gödel-Turing-OI incompleteness family operates at every level.

The framework's content on embedded computation is then organized by the conditional BQP model theorem stated in §4 below (with its extended development in [1, Ch. 14]): under the model hypotheses stated there, bounded-error quantum polynomial-time is the computational class accessible to an embedded observer. This is a conditional statement about the *ceiling* of embedded computation, not about its internal structure. On the internal structure — including the P vs NP question — the framework is silent in a precise sense developed here.

The paper's structure: §2 carries the structural chain from evolution to general intelligence, identifying the one contingent step (general intelligence emergence) and the structural inevitability of universal computation given that step. §3 develops the self-referential loop of AI as embedded-observer construction, with the partition-level limits applying to the constructed observer. §4 states the BQP characterization theorem and its consequences for the Extended Church-Turing Thesis and the conjectured BPP–BQP separation. §5 develops the framework's position on the P vs NP question — neutrality, structural silence, the unifying boundary-characterization pattern across the Gödel-Turing-OI incompleteness family, and the relationship to Aaronson's physical-principle approach. §6 examines why no obvious bridging axiom is available. §7 proposes three condition-based research directions for complexity-theoretic refinements within the BQP ceiling. §8 summarizes what the framework establishes and what it does not.

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2. From Evolution to Intelligence

Information processing is generically fitness-enhancing. In an environment with the derived physics — a thermal window [3, §2.7] guarantees fluctuating conditions ($kT > 0$) — organisms that respond to environmental information outcompete those that do not. A cell that detects a nutrient gradient and moves toward it outcompetes one that moves randomly. This is not a contingent fact about Earth — it follows from the structure: fluctuating environments reward better internal models of external conditions. Selection generically favors information processing.

Neural computation is structurally possible. Fast electrochemical signaling requires ions (Na^+, K^+, Ca^{2+} — present in the derived periodic table) moving through channels in lipid membranes. Carbon chemistry in d = 3 provides amphiphilic molecules (hydrophilic head + hydrophobic tail from the orbital structure) that self-assemble into bilayer membranes. The scale hierarchy [3, §2.5] separates neural signaling energies (~meV) from chemical bond energies (~eV), so information processing does not destroy the substrate.

General intelligence is structurally unconstrained but not guaranteed. The derived structure *permits* arbitrarily complex neural circuits (the orbital algebra provides materials, the scale hierarchy provides energetic separation), but whether evolution *reaches* general-purpose reasoning — abstraction, planning, language — depends on the specific fitness landscape. On Earth, general intelligence arose once in ~4 billion years. Whether this is evidence of rarity or inevitability the framework cannot determine. This is the one step in the chain where the structural argument is weakest: information processing is generically favored, but general intelligence is a much more specific capability.

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3. From Intelligence to AI and the Self-Referential Limit

The structural chain to universal computation. If general intelligence exists — for whatever contingent or structural reason — then AI follows from the derived structure through a specific chain in which every link is structural.

Band theory. The framework's emergent quantum representation [1, Part I] applied to electrons in a periodic crystal potential (the sp^3 diamond-cubic lattice of a Group IV element) gives Bloch's theorem: energy bands separated by gaps. The periodic potential is structural — it follows from sp^3 hybridization in d = 3. Band structure is a consequence of the framework's emergent quantum mechanics applied to derived crystal geometry.

Semiconductors. For Group IV elements (4 valence electrons per atom), the Pauli exclusion principle (derived — spin-statistics from staggered fermions [2, §4.2]) exactly fills the valence band and leaves the conduction band empty. The band gap for silicon (~1.1 eV) sits in the thermal sweet spot: $E_{\text{gap}}/kT \sim 40$ at room temperature. Large enough that thermal excitation doesn't flood the conduction band (the material isn't always conducting); small enough that a modest voltage promotes electrons across the gap (the material can be switched). The *existence* of Group IV semiconductors with gaps in the right range relative to the thermal window is structural — it follows from the scale hierarchy [3, §2.5]: the gap is set by the atomic energy scale ($\alpha^2 m_e c^2$), and the thermal window operates far below it.

Doping. Replacing a silicon atom (Group IV) with phosphorus (Group V, 5 valence electrons) adds one free electron — n-type. Replacing with boron (Group III, 3 valence electrons) removes one — p-type. Both dopant types exist in the derived periodic table. The ability to create n-type and p-type regions is a structural consequence of the periodic table architecture.

From transistors to universal computation. A p-n junction rectifies current. A transistor (two p-n junctions) amplifies and switches. Transistors in combination implement logic gates. A NAND gate is functionally complete — any Boolean function can be built from NAND gates alone. Sufficient NAND gates implement a universal Turing machine. Each step is either a structural consequence of semiconductor physics or a mathematical theorem about computation.

Step	Content	Status
Band theory	QM + periodic crystal potential	Structural
Band gap for Group IV	Exactly filled valence band (4 electrons + Pauli)	Structural
Semiconductor behavior	Gap in thermal sweet spot ($E_{\text{gap}}/kT \sim 40$)	Structural (scale hierarchy)
Doping	Group III / V substitution	Structural (periodic table)
p-n junctions	Interface between p-type and n-type	Structural consequence
Transistors	Voltage-controlled p-n junctions	Structural
Logic gates	Transistors in series / parallel	Structural
Universal computation	NAND is functionally complete	Mathematical theorem

The chain from (S, φ) to a universal computer is structural — given someone to build it. That is the same contingent step (general intelligence) identified in §2. The reconstruction theorem [5, §3] implies that a sufficiently sophisticated observer can reconstruct the local lattice/gauge invariants fixed by Theorem 23; it does not identify a unique Bell-inclusive microphysical completion. An observer can nevertheless exploit the reconstructed local physics to build computational devices.

The self-referential loop. AI is an embedded observer building another embedded observer within the same (S, φ). The characterization theorem [1, §3.4] applies to the new observer. It sees the same QM, the same $\hbar$, the same SM, the same dark sector. Not because it is limited by human design, but because the physics is structural — determined by the partition, not by the observer's complexity. A superintelligent AI has the same observational access as any other embedded observer: it reads the visible sector and writes to the hidden sector through the same coupling. It cannot determine h. It cannot see past the cosmological horizon. It can build better instruments, but it cannot overcome the partition.

The recursion — AI building AI building AI — produces ever more sophisticated embedded observers, each subject to the same structural limits. The Gödel-Turing-OI incompleteness family applies at every level: a formal system cannot prove its own consistency (Gödel), a computer cannot decide its own halting (Turing), an embedded observer cannot determine the hidden state (OI). No level of recursive self-improvement overcomes the partition. The limits are not technological — they are mathematical.

What AI can and cannot do.

Capability	Status
Reconstruct the local lattice/gauge invariants from observations	Structurally possible under Theorem 23's conditions
Build better models of the visible sector	No structural limit
Determine the hidden-sector state h	Provably impossible (characterization theorem)
Observe beyond the cosmological horizon	Provably impossible (causal partition)
Build faster computers	Structurally possible (derived physics permits it)
Replace a genuinely P-indivisible visible process by a faithful memoryless (divisible) model at the same horizon	Provably impossible when the measured memory witness is nonzero ([Main §3.4])
Recursive self-improvement	Possible within the structural limits

The framework's prediction is precise: there is no ceiling on the complexity of the observer, but there is a permanent floor on what any observer can access. The ceiling is contingent — it depends on the specific φ and the specific evolutionary and technological history. The floor is structural — it is the same for all embedded observers, biological or artificial, at any level of sophistication.

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4. A Conditional BQP Model Theorem

§3 establishes that embedded observers can construct universal computers from the framework's derived structure. The next question is whether there is a structural bound on the *complexity* of computation those constructed devices can achieve. The conditional BQP model theorem — stated here as the statement of record, with its extended development in [1, Ch. 14] — answers this *relative to an explicit computational model*, whose hypotheses are: a uniform efficient encoding of substratum configurations into qubits; efficient state preparation and control; a clock model; efficient measurement; explicit precision accounting; noise assumptions under which fault tolerance applies with polynomial overhead; and operational closure — all admissible operations available to the observer within the model are standard quantum instruments (CPTP maps for unconditioned evolution, CP trace-nonincreasing branches for conditioned outcomes), with no additional nonlinear, post-quantum, oracle, or hidden-sector-access operation, and with conditioned branches and postselected preparations charged their physical success-probability/repetition cost (no free postselection: PostBQP = PP). Under these hypotheses, bounded-error quantum polynomial-time is the computational class accessible to an embedded observer satisfying C1–C4; none of the hypotheses follows from CPTP representability of the visible dynamics alone, and each is a modelling commitment, not a consequence of C1–C4. The theorem combines a lower bound (BQP is available to any embedded observer satisfying C1–C4 under these hypotheses; the quantum-computing toolkit can be physically instantiated on the framework's substratum) with an upper bound (BQP cannot be exceeded under the same hypotheses; any super-BQP capability would falsify the conjunction of the framework's commitments with this model, sharpest at the model bridge).

4.1 The lower bound

Quantum computation in the standard model is defined through uniform polynomial-size circuit families: every BQP computation, by definition, is specified by such a family over a finite universal gate set — Hadamard, a non-Clifford phase gate such as T, and CNOT — with the Solovay–Kitaev theorem supplying efficient approximation of any fixed gate over that set. Generic n-qubit unitaries, by contrast, require circuits of size $\Theta(4^n)$ [11] and play no role in the definition. The lower-bound task is therefore to show that the framework's explicit computational model can realize the gates of a uniform polynomial-size family. Main supplies the transition-statistics layer — every finite law admits the fixed-basis unitary representation, and every finite adaptive experiment has a deterministic finite-test shadow [1, Part I] — while the controllability needed to realize chosen gates is the model's contribution: the gauge structure permits the qubit decomposition (a single SU(2) doublet provides the qubit basis), the lattice geometry permits the spatial layout, and §4's control hypotheses supply preparation and gating. Under those hypotheses, any BQP algorithm has a physical implementation on the substratum, and embedded observers can in principle access any problem in BQP.

4.2 The upper bound

No physical process implementable on the substratum exceeds BQP, under the model's efficiency hypotheses. The framework's dynamics is finite-rate (bounded coupling degree, A3 of the reconstruction [5, §3]), the partition entropy is finite (E3 plus the holographic interpretation), and the dynamics is unitary in the visible-sector marginal (by construction) — type-level facts that alone do not bound complexity; the load-bearing additions, matching [1, Ch. 14 §14.4], are that admissible evolutions are locally generated with uniformly computable descriptions, run for polynomial physical time at polynomial precision, and admit polynomial-overhead circuit simulation. Hypercomputation [6], oracle access [7], and other super-BQP capabilities would require either continuum precision (violating A1), unbounded coupling (violating A3), operations outside the operational-closure model (non-unitary CPTP evolution is ordinary open-system dynamics and stays inside; what is excluded is the nonlinear, post-quantum, oracle, or hidden-sector-access class), or violation of these efficiency hypotheses. The first falls outside the framework's class of admissible substrata; the operational-closure and efficiency violations fall outside the added computational model — a device exhibiting them would falsify global coverage of that model, not the substratum class by itself.

4.3 The combined result

Theorem (conditional BQP model). *Assume the computational model of §4: a uniform efficient encoding of substratum configurations into qubits; efficient state preparation and control; a clock model; efficient measurement; explicit precision accounting; noise assumptions under which fault tolerance applies with polynomial overhead; and operational closure — all admissible operations available to the observer within the model are standard quantum instruments (CPTP maps for unconditioned evolution, CP trace-nonincreasing branches for conditioned outcomes), with no additional nonlinear, post-quantum, oracle, or hidden-sector-access operation, and with conditioned branches and postselected preparations charged their physical success-probability/repetition cost (no free postselection:* PostBQP = PP*). Then for an embedded observer satisfying C1–C4, every computation the model can host lies in BQP (upper bound, §4.2), and every BQP computation can be hosted (lower bound, §4.1). CPTP representability of the visible dynamics implies none of the model hypotheses.*

Within the stated model, BQP is the *exact* ceiling — the class characterized by the framework's structural commitments *together with* the model hypotheses; outside the model the characterization is not claimed. This identifies the structural limit on embedded computation: there is no ceiling on the *complexity of the observer*, but there is a permanent ceiling on the *computational reach* of any embedded observer at BQP exactly.

4.4 The quantum (BQP-form) Extended Church-Turing Thesis as conditional theorem

The Extended Church-Turing Thesis (ECT) — that any physically realizable computation can be simulated by a probabilistic Turing machine with at most polynomial overhead — has long been treated as an empirical conjecture about the physical universe. The framework's conditional BQP model theorem yields the quantum (BQP-form) ECT — the BQP counterpart of the classical thesis — under the additional global-coverage premise — every physically realizable computation falling under the same operational-closure and efficiency hypotheses — as a conditional result rather than a theorem about unconditioned physical reality, in a modified form: the appropriate computational class is BQP rather than BPP, reflecting the quantum nature of the visible-sector marginal.

This modification has empirical consequences. The conjectured BPP–BQP separation — task-specific quantum advantage for problems like factoring, discrete log, and certain simulation problems — is reframed as the cost of partial observation rather than as evidence that the universe is fundamentally quantum. Under the standard separation conjecture, a BPP-limited classical implementation cannot reproduce all BQP computations efficiently; on the framework's reading, the hidden-sector correlations that enable the conjectured advantage are accessible only through measurement protocols that exploit the substratum's actual structure — quantum hardware exploits these correlations, while classical sampling does not. The observed task-specific quantum advantage — with the strict BPP–BQP separation conjectured, not proved — is read by the framework as evidence about the substratum's structure, not evidence against the framework.

4.5 The finite-substratum refinement: a ceiling above BQP's floor

The characterization of §4.3 is exact in the idealized limit $S \to \infty$. The framework, however, asserts that S is finite — the holographic bound caps the substratum resources available to any physical computation. This subsection states the finite-size consequence as a complexity-assumption-dependent proposal: its status inherits the standard conjectures made explicit below, and it is not entailed by Main.

The forced position. Any interpretation with a finite classical substratum faces a standard trilemma [7]: either (I) quantum computation encounters a scale ceiling; or (II) the tasks exhibiting quantum advantage possess hidden classical algorithms; or (III) the substratum commands resources exponential in the number of emergent qubits. The framework closes (III) by construction: the holographic bound is a structural commitment. It addresses (II) in one theorem-level respect: the substratum measure is the unique φ-invariant maximal-entropy measure ([Main], Lemma 3 and posit (iv) of the posit ledger), which excludes a planted-measure escape — though absence of planted answers is not itself a classical lower bound: the hardness of the relevant sampling rests on standard complexity-theoretic evidence (random-circuit results are assumption-dependent, e.g. non-collapse of the polynomial hierarchy), so (II) closes conditionally on those assumptions, while escape by conspiracy would additionally require a planted measure contradicting the uniqueness clause; independently, Theorem 24 of [Substratum §4] classifies the specific microstate as gauge, provably without empirical content, whereas extractable task answers would be empirical content. What remains, granted the complexity assumptions below, is (I): the ceiling is a conditional consequence with its assumptions stated, not an unconditional entailment.

The scale. The substratum budget for a computation of wall-time T is bounded by the holographic patch: $\mathrm{bits}(T) = \pi(cT)^2/l_p^2$ degrees of freedom updating at most once per t_p [10]. This gives $\sim 10^{145}$ operations for a day-scale computation, $\sim 10^{157}$ for thirty years, and $\sim 2 \times 10^{183}$ at the absolute horizon scale. A computation whose output statistics require classical sampling resources exceeding the budget cannot have those statistics realized by the substratum. The ceiling therefore sits, under an explicit quantitative assumption — $C_{\mathrm{cl}}(n, \varepsilon) \geq 2^{\alpha n}$ for the chosen circuit ensemble and fidelity criterion, an assumption the cited hardness evidence supports but does not prove — at $n^* \approx \log_2(\mathrm{budget})/\alpha$: with $\alpha = 1$, approximately 480 (day), 520 (decades), 610 (absolute) *full-fidelity, sampling-hard* logical qubits — the operative quantity being fidelity-weighted classical simulation cost, not qubit count.

What survives and what fails. The binding variable is the classical cost of the specific task. Factoring is subexponential classically (RSA-2048 $\sim 2^{117}$, RSA-4096 $\sim 2^{156}$, both far inside budget): Shor's algorithm at all cryptographically relevant sizes is unaffected, with the crossover near $\sim$ 86-kilobit keys. Clifford circuits are classically simulable and hence unconstrained at any scale. What the ceiling forbids is full-fidelity realization of genuinely exponential-cost tasks — random-circuit sampling and generic highly entangled simulation — beyond $n^* \sim 500$–600.

The signature. The prediction is qualitatively unlike any conventional noise model: fidelity should track the classical simulation cost of the circuit rather than its size, with degradation onsetting in (for example) T-count beyond the budget scale while structurally comparable Clifford-dominated circuits remain clean. Hardware, on the standard account, is blind to circuit semantics; hardness-correlated failure at fixed gate quality is therefore a discriminating signature. It is separable from the non-Markovian per-gate correction of [Complexity §8.2], which is task-independent bath physics: the two effects constitute distinct walls with distinct scalings.

Falsification. Current experiments sit roughly 390 doublings below the ceiling (fidelity-weighted hardness $\sim 2^{95}$ against a day-scale budget of 2^{482}); nothing yet probes it. Fault-tolerant machines at $\sim 10^3$ logical qubits enter the test zone. Clean absence of hardness-correlated degradation on sampling-hard circuits at that scale — full-fidelity performance where the budget forbids it — falsifies, in order of exposure, the quantitative cost model (the α-assumption below), then the complexity-resource bridge — and only if those survive every repair does the pressure reach the finite classical substratum and [Main].

Corollary (entanglement dimension). The same budget caps the entanglement dimension available to any physical Bell-type experiment at $2^{S_{\text{patch}}}$: correlations requiring strictly infinite-dimensional (commuting-operator) strategies, whose existence is implied by MIP*=RE, are physically unrealizable in-framework. The framework thereby takes a definite side on the physical Tsirelson question: realizable correlations are those approximable by finite-dimensional strategies.

Conditions. The statement is conditional on: standard complexity conjectures (no polynomial classical algorithms for the sampling tasks; non-collapse of the polynomial hierarchy); the declared posits (iv)–(v) of the ledger ([Main]), which are what certify the absence of planted structure; the budget arithmetic at order-of-magnitude precision (the exponent is robust; prefactors are not); and the full-fidelity qualification, since the constraint binds fidelity-weighted cost, so low-fidelity demonstrations do not approach it.

4.6 The depth axis: residual dissipation, the correction fork, and the box

§4.5 bounded the width of physical computation; the framework also bounds its depth. The residual dissipation $\mathcal{D} \sim (\tau_S/\tau_B)^2 \sim 10^{-64}$ ([Main], §3.2 and the scope remarks of §3.5) is an isolation-independent deviation from exact unitarity sourced by the cosmological partition itself — a decoherence floor in the Penrose–Diósi genre but with a derived rate, already registered in [Main] as the framework's falsifiable residue. Its computational reading: uncorrected coherent depth is bounded by $\sim 1/\mathcal{D} \sim 10^{64}$ operations at the system's natural clock, however perfect the engineering.

The correction fork, resolved. Whether quantum error correction erases this wall is determined, not open. The $\mathcal{D}$-channel is quasi-static — its conditioned Hamiltonian is static over the computation (correlation time $\sim \tau_B$, the Hubble-scale bath memory) — so it presents as a temporally static, largely common-mode coherent error: the benign class for fault tolerance, handled by calibration, dynamical decoupling, and the error correction of low-weight coherent errors, and some sixty orders below threshold besides. The channel therefore satisfies the independence assumptions of the threshold theorem, logical depth is unbounded under error correction, and the $\sim 10^{64}$-operation cap is a property of *uncorrected* hardware only. This resolves the fork toward erasure — an honest weakening of the box's depth face, recorded with the same prominence as the framework's strengthenings: the depth wall is not a fault-tolerant limit.

The box. With this section the framework bounds all four axes of physical computation from one construction: width (holographic budget, §4.5), depth (residual dissipation, this section), rate (the Margolus–Levitin bound, derivable from the tick structure — derivation queued), and memory (the Bekenstein bound, derivable from the partition — derivation queued). Hypercomputation through spacetime structure is closed as well: Malament–Hogarth constructions require unbounded blueshift and effectively infinite state space, unavailable on a finite lattice with e^S states. The account of physical computation is therefore a four-axis program: the energy and spacetime faces carry derived exclusions, the width face a complexity-assumption-dependent proposal (§4.5), and the rate and memory faces queued derivations from the same primitives — a program, not a completed box.

Near-term texture. The same partition machinery predicts P-indivisible device noise at the engineered-bath level ([Complexity §8.2]); its syndrome-level signature — an exploitable temporal kernel in detection-event time series, distinguishable from classical drift by divisibility tests — is testable on already-public fault-tolerance datasets, and the sharpened prediction is stated there.

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5. The Internal Structure of BQP and the P vs NP Question

The BQP characterization theorem identifies the *boundary* of computational capability for embedded observers. It is silent on the internal structure of complexity classes inside that boundary. The conventional inclusion chain P $\subseteq$ NP $\subseteq$ PH $\subseteq$ PSPACE, with BQP positioned somewhere within PSPACE and conjectured to contain BPP strictly, is unmodified by the framework's content.

5.1 The location of P vs NP within the framework

The framework's conditional BQP characterization places P inside BQP (P $\subseteq$ BPP $\subseteq$ BQP); the relation of NP to BQP is UNKNOWN — efficient witness verification puts the verification map, not the NP language, inside BQP, and NP $\subseteq$ BQP is an open containment question. Consequently the framework's conditional ceiling neither settles P vs NP nor determines the NP–BQP relationship. The conventional inclusion chain is unmodified by the framework's content. The framework's contribution is at a different level: it identifies BQP as the *outermost* class accessible to embedded observers, leaving the inclusion relationships among interior classes — including P vs NP — to standard complexity theory.

5.2 The neutrality result

The framework's commitments do not entail either P = NP or P $\neq$ NP. Two cases:

Case 1: $P \neq NP$ (the conventional expectation). P is inside BQP while the NP–BQP relation remains open; no efficient general solution is known for NP-complete problems; quantum advantage (Shor's algorithm and related results) places specific NP problems inside BQP without placing all of NP inside either BQP or P, consistent with the standard conjectures.

Case 2: $P = NP$ (the surprising case). A polynomial-time algorithm for an NP-complete problem operates on a classical Turing machine; its outputs are in P, and by inclusion in BQP. The framework's BQP ceiling is unchanged: the new algorithm provides additional computational reach for embedded observers but does not exceed BQP. The framework would not be falsified by this discovery.

Both cases sit inside the framework's BQP ceiling. The framework's empirical exposure on P vs NP is zero: no observation an embedded observer could make would distinguish the two cases at the level of the framework's commitments.

5.3 Relation to gauge vs computational incompleteness

The framework's Methodology paper [8, §19] distinguishes *gauge incompleteness* (the question has no answer-bearing content; no observation could distinguish) from *computational incompleteness* (a definite fact exists but is unreachable through finite observation). The distinction bears on P vs NP carefully.

P vs NP is a determinate mathematical question with a definite answer — the abstract fact about whether a polynomial-time algorithm for SAT exists is either true or false, even though we do not know which. The question is not vacuous in the gauge sense (it has answer-bearing content). What the framework's content adds is that the determinate mathematical answer, whichever it is, is *physics-invariant* relative to the framework: the same empirical content (BQP as the ceiling) is consistent with both answers. This is weaker than gauge — not "no fact of the matter" but "the fact carries no consequence for the framework's empirical claims." The framework is uncommitted on P vs NP, and a mathematical proof of either answer would leave the framework's empirical predictions unchanged.

5.4 The silence is structural, not stipulative

The neutrality result of §5.2 can be sharpened. The framework's two foundational axioms (tokened differentiation occurs; differentiation recurs) [8] and their derived structural commitments — finiteness-per-moment, bijectivity, the proper-part decomposition $V \subsetneq S$, the C1–C4 partition conditions — deliver content of one specific type: they characterize the *boundary structure* between substratum and embedded observer, and thereby characterize the computational ceiling at that boundary. They do not deliver content of the type required to bear on internal complexity-class structure.

P vs NP is a question about the existence of polynomial-time algorithms for NP-complete problem families, defined asymptotically over inputs of unbounded size. The framework's axiomatic content concerns finite physical structures (with possibly unbounded history but always finite at any moment). The bridge between these two domains has a structural gap: physical-substrate axioms about finite structures do not entail asymptotic statements about algorithmic complexity classes without an additional bridging premise that the axioms themselves do not provide.

The framework's silence on P vs NP is therefore not a chosen agnosticism nor a residual of incomplete development — it is a feature of what kind of question P vs NP is, relative to what kind of content the framework's axioms deliver. A foundational program whose axioms *did* constrain P vs NP would require axioms of a different type, bridging physical substrate to asymptotic algorithmic structure directly.

A positive complement to this silence is worth noting briefly. The framework's finiteness commitment — that the substratum has finitely many tokens at any moment — sits naturally with *resource-bounded* complexity theory (specific input sizes, specific resource bounds) rather than with asymptotic complexity theory (limits as input size $n \to \infty$). Asymptotic complexity classes are mathematical idealizations defined over unbounded input families; their physical referent in a finite substratum is necessarily indirect. Resource-bounded complexity, by contrast, asks for inputs of specific bounded size what the cost of solving specific problems is — closer to actual computational practice, since real computers solve specific instances rather than asymptotic families. The framework is silent on asymptotic P vs NP because the question lives in the mathematical access mode (Methodology §17 in its sharpened form); within the physical access mode the framework natively addresses, the relevant complexity questions may be more fruitfully posed as resource-bounded questions about specific embedded-observer architectures. This is not a refutation of asymptotic complexity theory — asymptotic theory is the natural setting for the mathematical access mode — but it identifies where the framework's own complexity content might productively engage with complexity theory: in fine-grained / parameterized / resource-bounded complexity rather than at the asymptotic P vs NP boundary.

5.5 The boundary-characterization pattern across the Gödel-Turing-OI family

The framework's silence on P vs NP fits a broader pattern across the incompleteness family identified in [1, Ch. 12 §12.8]: Gödel's incompleteness, Turing's halting result, OI's hidden-sector inaccessibility, and Deutsch's universality-of-explanation are all *boundary-characterization* results that share a common structural feature — each establishes a structural limit on access from inside a system, and each is silent on the internal organization of what falls within the accessible domain.

Concretely:

- **Gödel** sets a boundary at *provability from within a consistent recursively-axiomatizable formal system.* Gödel's incompleteness statements are about what crosses that boundary (true statements that do not). They are silent on the internal organization of the provable: which provable statements are interesting, useful, or important is not a question Gödel's theorems address.

- **Turing** sets a boundary at *computability by a Turing machine.* The halting problem statement is about what crosses that boundary (instances whose halting cannot be decided). Turing's theorems are silent on the internal organization of the computable: which decidable problems are tractable, which are intractable, is not a question Turing's theorems answer.

- **OI** sets a boundary at *embedded observability of the substratum* (the hidden-sector state h, the cosmological horizon, the substratum representative). The OI incompleteness statements are about what crosses that boundary. They are silent on the internal organization of the observable: which physical phenomena have particular structure within the framework's accessible domain is content of a different kind, addressed by the framework's specific structural commitments (gauge group, generations, etc.) rather than by the incompleteness statement itself.

- **Deutsch** sets a boundary at *universal explanation* in the constructor-theoretic sense. Deutsch's universality-of-explanation is about what crosses that boundary (explanations attainable by sufficient knowledge-creation). It is silent on which problems within explanatory reach get solved in which order, by which methods.

This shared structure — establishing a boundary, then silence on the interior — is not an accident of how the four results have been developed. It reflects a common feature of structural-incompleteness frameworks: each identifies a *mode of access* (provability, computability, observability, explanation) and characterizes its limits. The internal structure of what falls within that mode of access is a different question, requiring tools internal to that domain.

P vs NP lies in the internal structure of the computable (within Turing's domain) and in the internal structure of BQP (within OI's domain). It is therefore outside the locus of contribution of any of these incompleteness frameworks. This is a constraint on what structural-incompleteness frameworks can and cannot deliver: they characterize boundaries; they do not navigate interiors. Internal-complexity questions like P vs NP require complexity-theoretic tools (proof-system lower bounds, geometric complexity theory, circuit complexity, and similar), not incompleteness frameworks.

The OI framework's silence on P vs NP is therefore *not* an idiosyncrasy of OI — it is the same kind of silence Gödel and Turing exhibit on the same question, for the same structural reason. The unifying observation: structural-incompleteness frameworks characterize the *boundary* of a domain of access; the *internal complexity* of what's accessible is a separate question requiring tools internal to the relevant complexity theory.

5.6 Position relative to Aaronson's physical-principle approach

Aaronson [4] has argued that the physical universe respects the principle that NP-complete problems are not solvable in polynomial time on physically realizable hardware — that P $\neq$ NP holds with the status of a physical principle. The OI framework is stronger and weaker than this position in different directions:

- **Stronger:** the framework commits to BQP exactly as the embedded-observer ceiling, where Aaronson's principle leaves the ceiling between P and BQP underdetermined. The framework specifies the boundary; Aaronson's principle does not.
- **Weaker:** the framework declines to commit to P $\neq$ NP itself, where Aaronson's principle entails it. Aaronson posits P $\neq$ NP as a physical principle rather than deriving it from physical axioms, because the latter cannot deliver it; the OI framework's structural silence on P vs NP is the principled version of this same observation.

The two positions are complementary rather than competing. Aaronson's physical-principle approach captures an intuition about physical realizability that is consistent with the framework's BQP ceiling but goes further by stipulating the answer to P vs NP. The framework declines to stipulate, on grounds that the stipulation is not derivable from physical-substrate axioms.

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6. Bridging Axioms: Why None Are Obviously Available

It is natural to ask what additional axiom would close the gap between physical-substrate content and asymptotic complexity-class content. Several candidate forms are available, each illustrating the structural difficulty:

A computational-irreducibility axiom asserting that most NP-complete instances admit no shortcut faster than brute-force enumeration would force P $\neq$ NP if made precise. The precise version, however, tends either to reduce to a complexity-class statement (rendering the axiom circular) or to lose its bite.

A computational-manifestation axiom asserting that complexity classes have physical instantiations would force conclusions about P vs NP only if combined with further constraints on what is physically realizable. Those constraints are essentially the question itself in disguise.

A direct bridging axiom positing a bound on physically realizable computation — e.g., "all physical computation is in P" or "no physical process solves NP-complete problems in polynomial time" — would settle P vs NP by stipulation rather than derivation. This is the structure of Aaronson's physical-principle move, and it is posited as a principle rather than derived from physics axioms precisely because the latter cannot deliver it.

The structural feature common to these candidates is that the bridge from "finite physical structure at each moment" to "asymptotic algorithmic complexity over all input sizes" requires content that the standard repertoire of physical axioms does not contain. Whether a principled bridging axiom exists — one that is independently motivated rather than question-begging, and that does not collapse to a known open conjecture — is itself an open methodological question. The framework's stance is that the BQP characterization is the legitimate physics-principled result within reach of physical-substrate axioms, and that internal complexity-class structure (including P vs NP) is the domain of mathematical complexity theory rather than of physical foundations.

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7. Beyond Axioms: Conditions and Framework-Relative Refinements

The previous section's negative result — that physical-substrate axioms do not bridge to asymptotic complexity content — should not be read as foreclosing all framework-relative complexity-theoretic content. The framework's structural commitments suggest content not at the *universal* P vs NP boundary but at framework-relative *conditional* boundaries within the BQP ceiling. These are research directions where the framework's specific substratum structure (cubic lattice, gauge group, partition structure) does provide content.

We sketch three such directions, each of which is a framework-relative refinement rather than a universal complexity-theoretic claim. The framework predicts none of them as new theorems; each is offered as a structurally motivated conjecture inviting complexity-theoretic investigation.

7.1 Geometric structure on the substratum lattice

NP-complete problems vary in the extent to which their constraint structure can be embedded into the substratum's cubic lattice geometry. Problems with locally-bounded constraint structure on cubic-lattice-embeddable graphs (planar 3-SAT, lattice-Ising models, certain graph problems) admit natural substratum representations; problems whose constraints require highly nonlocal couplings do not. The framework's conjecture in this direction is that lattice-local NP-complete problems admit *physical solution protocols* whose complexity reflects the locality structure — specifically, that physical implementation of such protocols incurs no overhead beyond the local-constraint structure, while nonlocal protocols incur overhead proportional to the nonlocality. This is not P vs NP; it is a refinement of how *implementation cost* on the framework's substratum varies with problem geometry. It connects to existing complexity-theoretic work on planar variants of NP-complete problems and on local Hamiltonian complexity [9].

7.2 Task-specific quantum advantage, hidden group structure, and the SU(2)-compatibility refinement

Existing complexity-theoretic evidence suggests that task-specific quantum advantage is strongest for problems with strong global structure (factoring, discrete log, hidden-subgroup problems) and smallest for problems with primarily local structure (where classical heuristics often perform well). The framework's partition architecture suggests a structural reason for this correlation: BQP advantage derives from exploiting hidden-sector correlations that are induced by nonlocal substratum couplings, and nonlocal substratum couplings are precisely what global problem structure requires. The base conjecture is that task-specific quantum advantage correlates structurally with the *globality* of the problem's hidden-sector dependence.

A sharper structural reading is available when the conjecture is restricted to its appropriate scope and tested against the catalog of known BQP advantages.

Scope restriction. The conjecture as stated applies to *decision and search* problems with identifiable group-theoretic hidden structure — the hidden-subgroup problem (HSP) family and its relatives. Sampling problems (boson sampling, random circuit sampling) constitute a structurally distinct class whose advantage derives from sampling-vs-decision separations rather than from extracting hidden group properties; the framework's structural reasoning does not directly engage them, and they are left outside the conjecture's scope. Within the in-scope class, the conjecture's testable content survives.

Survey against the known advantage catalog. The known task-specific quantum advantages with identifiable group-theoretic structure span several distinct epistemic types — proven query-model separations, polynomial-time quantum algorithms with conjectured classical hardness, BQP-complete estimation problems, and subexponential speedups — and cluster as follows: Shor's factoring (HSP on $\mathbb{Z}_n^*$, abelian; a polynomial-time quantum algorithm with classical hardness conjectured, not proved), discrete log (HSP on cyclic groups, abelian; the same status), period-finding (HSP on $\mathbb{Z}$; the same status), Simon's problem (HSP on $(\mathbb{Z}_2)^n$, abelian; a proven exponential query-model separation), Deutsch-Jozsa ($(\mathbb{Z}_2)^n$; a query-model result only — a bounded-error randomized classical algorithm needs just constantly many queries, so it witnesses no exponential bounded-error advantage), Forrelation (Fourier-correlation on $(\mathbb{Z}_2)^n$; a proven query-model separation, not an unrelativized BPP $\neq$ BQP), and Jones polynomial estimation at roots of unity (Aharonov-Jones-Landau 2006, a BQP-complete estimation problem, with hidden structure in the braid-group representations factoring through $\mathrm{SU}(2)_q$). On the negative side, large symmetric groups S_n and dihedral groups D_n for large n — both nonabelian with complex representation theory — have no known polynomial-time quantum algorithm despite extensive investigation — for the dihedral case Kuperberg's algorithm runs in subexponential $2^{O(\sqrt{\log N})}$ time [12], a subexponential speedup rather than an exponential advantage; this is the empirical state of the nonabelian HSP problem.

The SU(2)-compatibility refinement. The structural reason behind this clustering becomes visible when one notes that the observer's gate toolkit — the framework's SU(2) representation structure with model-supplied controllability (Chapter 1 of [1] gives the representation; §4's hypotheses the control) — is built from SU(2)-based operations: qubits are SU(2)-doublet representations; the universal gate set (Hadamard, T, CNOT) acts on these doublets. The quantum Fourier transform — the workhorse of HSP-style algorithms — is efficiently implementable for groups whose representation theory factors through SU(2)-compatible structure: abelian groups embedding in U(1) (Shor, discrete log, period-finding), elementary 2-groups via SU(2)-derived $\mathbb{Z}_2$ structure (Simon, Deutsch-Jozsa, Forrelation), small symmetric groups via low-dimensional $\mathrm{SU}(n)$ embeddings, tractable nonabelian groups (affine, Heisenberg) with abelian-normal structure, and groups with quantum deformations of SU(2) representation theory (Jones polynomial via $\mathrm{SU}(2)_q$). Groups whose representation theory does not factor through SU(2)-compatible structure — large S_n, large D_n, generic non-Lie combinatorial groups — appear to be precisely the cases where exponential quantum advantage is elusive.

Refined conjecture (v5): *Within the class of decision and search problems with identifiable group-theoretic hidden structure, exponential BQP advantage correlates with whether the hidden group's representation theory is* SU(2)*-compatible — that is, factors through* SU(2) *representations or their quantum deformations in a way that the framework's* SU(2)*-based quantum toolkit can efficiently access. The framework's broader gauge group* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *determines what physics is; the BQP-advantage correlation connects specifically to* SU(2) *via the* SU(2) *representation structure on which the model's toolkit is built.*

This is the structural reading of the task-specific-advantage pattern that the framework's content suggests. It survives the survey against known BQP advantages (twelve cases, all in-scope cases consistent), accommodates the tractable / intractable boundary in nonabelian HSP, and is grounded in the framework's existing QM emergence theorem rather than being an external postulate. The conjecture is silent on sampling-style advantages (boson sampling, random circuit sampling) by scope, not because the framework is inconsistent with them. The internal investigation supporting this refinement is recorded in the framework's working archive (record §B.2.191).

Three concrete checks remain testable. (1) The known exponential-BQP-advantage decision/search problems should all admit hidden structure factoring through SU(2)-compatible representation theory. (2) The tractable-vs-intractable boundary in nonabelian HSP should correspond to whether the hidden group has SU(2)-compatible representation theory or doesn't. (3) New BQP-advantage decision/search problems should be most discoverable by looking for hidden group structures with SU(2)-compatible representation theory, including quantum-deformation variants. None of these is established by the framework; each is a structurally motivated conjecture inviting complexity-theoretic investigation. The framework's contribution is to suggest a specific structural reason — the SU(2)-based, model-supplied toolkit construction — behind the empirical concentration of BQP advantage on the specific group structures it concentrates on, with this contribution rooted in the framework's existing QM emergence theorem rather than introduced as new content.

7.3 Gauge equivalence and complexity-theoretic reductions

The framework's substratum gauge group $\mathscr{G}_{\mathrm{sub}}$ partitions the space of bijections into equivalence classes; the gauge group includes state relabeling, alphabet change, deep-sector enlargement, and graph isomorphism up to statistical isotropy (cubic-structure-preserving for the gauge sector; [5, §4, Theorem 24(iv)]). A natural question is whether $\mathscr{G}_{\mathrm{sub}}$ has an image in the space of polynomial-time reductions among NP-complete problems. If gauge equivalence at the substratum level corresponds (in some precise sense) to polynomial-time reducibility at the algorithm level, the framework's gauge structure would provide a *physical* account of why NP-complete problems are reducible to each other. This is speculative — the correspondence has not been made precise — but the structural similarity is suggestive: gauge equivalence is "no observable distinction" at the substratum level, and polynomial-time reducibility is "no complexity distinction" at the algorithm level. Both are equivalence relations on otherwise distinct structures.

7.4 Conditions and the right kind of research question

The pattern across the three directions in §7.1–§7.3 is that the framework's specific structure suggests *conditional* complexity-theoretic claims — claims of the form "for problems satisfying condition X (lattice-locality, global structure, gauge-equivalence to another problem), the framework's structure suggests a specific complexity-theoretic behavior." This is the appropriate level at which the framework engages complexity theory: not by attempting to bridge to universal questions like P vs NP, but by suggesting how the framework's specific substratum architecture refines the internal landscape of BQP.

These conditional refinements are not predictions of the framework in the sense of the SM observable predictions [2, §7]; they are research directions opened by the framework's substratum structure. Some may be productive, others may not. The framework's contribution is to identify them as natural questions given its structure, not to resolve them.

8. What the Framework Establishes and What It Does Not

Summarizing the paper's content:

The framework establishes (theorems):

- Under the explicit computational model of §4, BQP is the computational class accessible to embedded observers satisfying C1–C4 (conditional BQP model theorem, §4; extended development in [1, Ch. 14]).
- The Extended Church-Turing Thesis (in BQP form) is a conditional theorem under the global-coverage premise together with §4's computational model, the coverage carrying the universal quantifier — not an empirical conjecture, not a bare consequence of C1–C4, and not of embeddedness.
- The conjectured BPP–BQP separation is structural in the framework's reading — the cost of marginal-statistics observation rather than evidence for fundamental quantum-ness.
- The Gödel-Turing-OI incompleteness family operates at every level of recursive self-improvement; no level of AI sophistication overcomes the partition.

The framework does not establish (silences):

- The internal structure of complexity classes inside BQP. P vs NP, in particular, is neither implied nor opposed by the framework's content.
- Universal complexity-theoretic claims that would require bridging finite-physical-structure axioms to asymptotic-algorithmic content. The bridge does not appear achievable from physical-substrate axioms alone.

The framework opens (research directions):

- Geometric refinements based on substratum lattice structure (§7.1).
- task-specific quantum-advantage structure correlating with problem globality (§7.2).
- Gauge equivalence as a possible physical account of polynomial-time reducibility (§7.3).

The division of labor is the appropriate one between the framework and complexity theory. The framework's content is a structural theorem about the computational ceiling for embedded observers and a set of research directions opened by the substratum's specific structure; standard complexity theory's content is about the internal structure of classes inside that ceiling. The two are complementary rather than competing.

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References

[1] A. Maybaum, *The Incompleteness of Observation*, (2026).

[2] A. Maybaum, "The Standard Model from a Cubic Lattice," (2026).

[3] A. Maybaum, "Structural Consequences of Observational Incompleteness: From Chemistry to Evolution," (2026).

[4] S. Aaronson, "NP-complete problems and physical reality," *ACM SIGACT News* **36**, 30 (2005); arXiv:quant-ph/0502072.

[5] A. Maybaum, "The Substratum Construction: Reconstruction, the Substratum Gauge Group, and the Synthesis of Quantum Mechanics with General Relativity," (2026).

[6] M. Davis, "The myth of hypercomputation," in *Alan Turing: Life and Legacy of a Great Thinker*, ed. C. Teuscher (Springer, 2004), pp. 195–211.

[7] S. Aaronson, *Quantum Computing Since Democritus* (Cambridge University Press, 2013).

[8] A. Maybaum, "Methodology and Foundations of the Incompleteness of Observation Framework," (2026).

[9] J. Kempe, A. Kitaev, and O. Regev, "The complexity of the local Hamiltonian problem," *SIAM Journal on Computing* **35**, 1070 (2006); arXiv:quant-ph/0406180.

[10] S. Lloyd, "Computational capacity of the universe," *Phys. Rev. Lett.* **88**, 237901 (2002); arXiv:quant-ph/0110141.

[11] V. V. Shende, S. S. Bullock, and I. L. Markov, "Synthesis of quantum-logic circuits," *IEEE Trans. Computer-Aided Design* **25**, 1000 (2006); arXiv:quant-ph/0406176.

[12] G. Kuperberg, "A subexponential-time quantum algorithm for the dihedral hidden subgroup problem," *SIAM J. Comput.* **35**, 170 (2005); arXiv:quant-ph/0302112.